

EDUQUALITY ACADEMIC FOUNDATION

CENTER FOR ACADEMIC EXCELLENCE & UN SDG 4 QUALITY EDUCATION

"Quality Education – Learn Today, Build a Better Tomorrow."

MATHEMATICS & STATISTICS

Advanced Mathematics & Inferential Statistics

Analytical Foundations, Differential Forms, Probability Spaces, and
Hypothesis Testing

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This textbook has been compiled by university professors and pedagogical researchers to support deep conceptual comprehension and self-paced mastery. Use the structured review prompts at the end of each module.

CHAPTER 1

Vector Spaces and Linear Transformations

1.1 Linear Independence, Bases, and Linear Mappings

Let V be a vector space over field F . A set of vectors $\{v_1, \dots, v_k\}$ is linearly independent if $c_1 v_1 + \dots + c_k v_k = 0$ implies all $c_i = 0$. The rank-nullity theorem proves $\dim(V) = \dim(\ker T) + \dim(\text{im } T)$ for any linear map $T: V \rightarrow W$. Matrix diagonalization $A = P D P^{-1}$ simplifies recursive systems and spectral analysis.

Key Conceptual Takeaways:

- Dimension of a vector space is the cardinality of any basis.
- Invertible linear transformations preserve algebraic structure and dimensionality.

CHAPTER 2

Probability Measure Spaces & Inferential Statistics

2.1 Axiomatic Probability, Random Variables, and Limit Theorems

A probability space is a triple $(\Omega, \mathcal{F}, P)$ satisfying Kolmogorov axioms. The Central Limit Theorem establishes that the normalized sum of independent and identically distributed random variables with finite variance converges in distribution to a standard normal distribution $N(0, 1)$, justifying parametric statistical tests.

Key Conceptual Takeaways:

- Expected value and variance quantify central tendency and dispersion of distributions.
- p-value measures the probability of observing data at least as extreme under the null hypothesis.

Selected Bibliography & Recommended Literature

- [1] UNESCO (2025). "Global Education Monitoring Report: Inclusion and Quality Benchmarks."
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- [5] Raworth, K. (2017). "Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist." Chelsea Green Publishing.

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